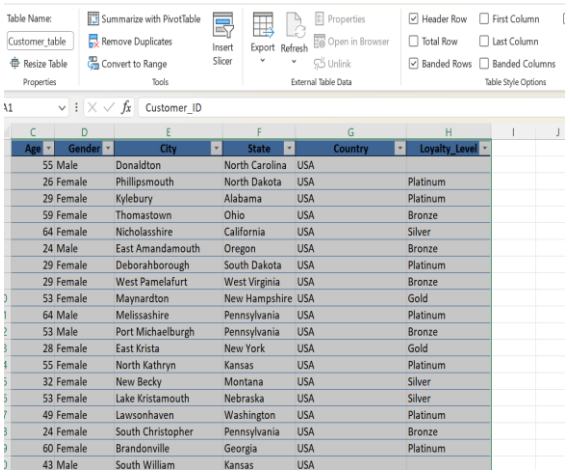
Module end assignment :

1.Data entry, import, and basic organization using tables

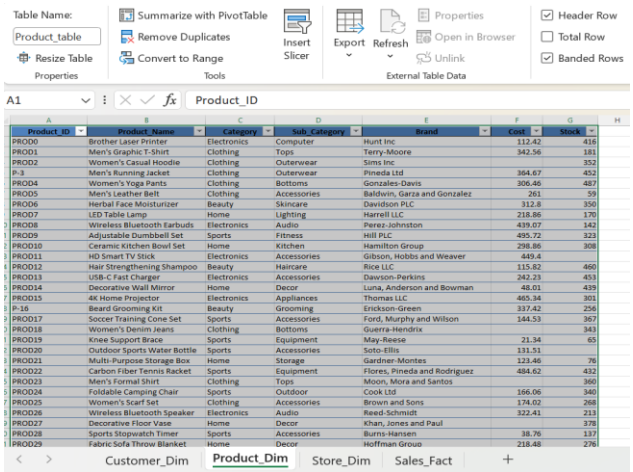
Converting all the sheets into tables and giving names

Customer_table



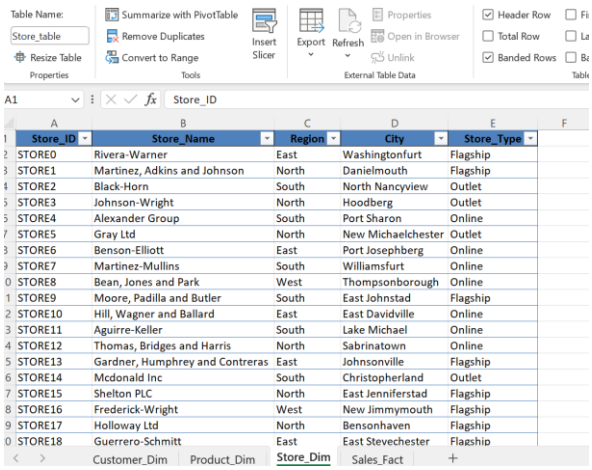
Age	Gender	City	State	Country	Loyalty_Level
55	Male	Donaldton	North Carolina	USA	
26	Female	Phillipsmouth	North Dakota	USA	Platinum
29	Female	Kylebury	Alabama	USA	Platinum
59	Female	Thomastown	Ohio	USA	Bronze
64	Female	Nicholasshire	California	USA	Silver
24	Male	East Amandamouth	Oregon	USA	Bronze
29	Female	Deborahborough	South Dakota	USA	Platinum
29	Female	West Pamelafurt	West Virginia	USA	Bronze
53	Female	Maynardton	New Hampshire	USA	Gold
64	Male	Melissashire	Pennsylvania	USA	Platinum
53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
28	Female	East Krista	New York	USA	Gold
55	Female	North Kathryn	Kansas	USA	Platinum
32	Female	New Becky	Montana	USA	Silver
53	Female	Lake Kristamouth	Nebraska	USA	Silver
49	Female	Lawsonhaven	Washington	USA	Platinum
24	Female	South Christopher	Pennsylvania	USA	Bronze
60	Female	Brandonville	Georgia	USA	Platinum
43	Male	South William	Kansas	USA	

Product_table



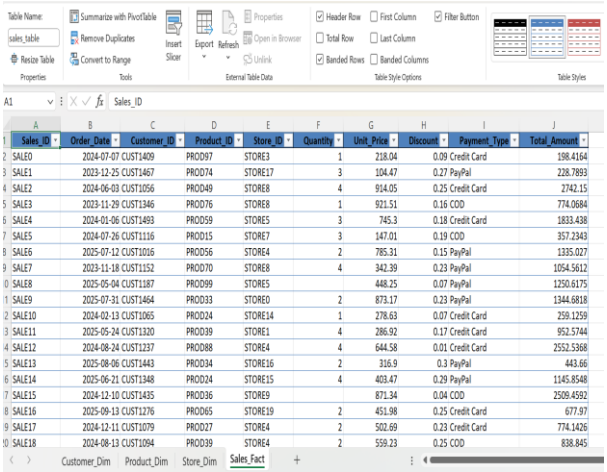
Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc		352
P-3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452
PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.8	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4	
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowman	48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301
P-16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix		343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51	
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodriguez	484.62	432
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos		360
PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340
PROD25	Women's Scarf Set	Clothing	Accessories	Brown and Sons	174.02	268
PROD26	Wireless Bluetooth Speaker	Electronics	Audio	Reed-Schmidt	322.41	213
PROD27	Decorative Floor Vase	Home	Decor	Khan, Jones and Paul		378
PROD28	Sports Stopwatch Timer	Sports	Accessories	Burns-Hansen	38.76	137
PROD29	Fabric Sofa Throw Blanket	Home	Decor	Hoffman Group	218.48	276

Store_table



Store_ID	Store_Name	Region	City	Store_Type
STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
STORE1	Martinez, Adkins and Johnson	North	Danielmouth	Flagship
STORE2	Black-Horn	South	North Nancyview	Outlet
STORE3	Johnson-Wright	North	Hoodberg	Outlet
STORE4	Alexander Group	South	Port Sharon	Online
STORE5	Gray Ltd	North	New Michaelchester	Outlet
STORE6	Benson-Elliott	East	Port Josephberg	Online
STORE7	Martinez-Mullins	South	Williamsfurt	Online
STORE8	Bean, Jones and Park	West	Thompsonborough	Online
STORE9	Moore, Padilla and Butler	South	East Johnstad	Flagship
STORE10	Hill, Wagner and Ballard	East	East Davidville	Online
STORE11	Aguirre-Keller	South	Lake Michael	Online
STORE12	Thomas, Bridges and Harris	North	Sabrinatown	Online
STORE13	Gardner, Humphrey and Contreras	East	Johnsonville	Flagship
STORE14	Mcdonald Inc	South	Christopherland	Outlet
STORE15	Shelton PLC	North	East Jenniferstad	Flagship
STORE16	Frederick-Wright	West	New Jimmymouth	Flagship
STORE17	Holloway Ltd	North	Bensonhaven	Flagship
STORE18	Guerrero-Schmitt	East	East Stevechester	Flagship

sales_table

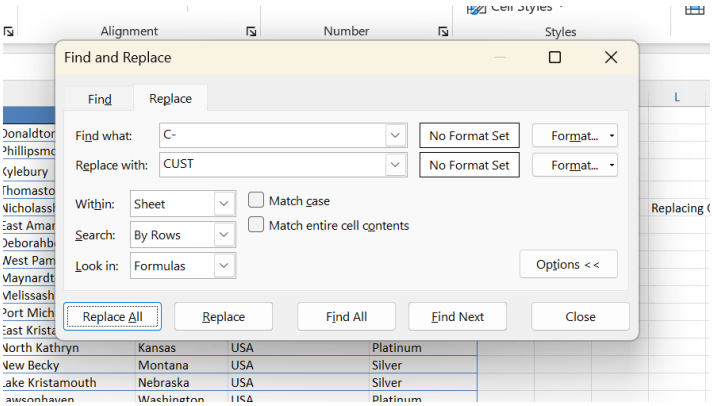


Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	2024-07-07	CUST1409	PROD97	STORE3	1	218.04	0.09	Credit Card	198.4164
SALE1	2023-12-25	CUST1467	PROD74	STORE17	3	104.47	0.27	PayPal	228.7893
SALE2	2024-06-03	CUST1056	PROD49	STORE8	4	914.05	0.25	Credit Card	2742.15
SALE3	2023-11-29	CUST1346	PROD76	STORE8	1	921.51	0.16	COD	774.0684
SALE4	2024-01-06	CUST1493	PROD59	STORE5	3	745.3	0.18	Credit Card	1833.438
SALE5	2024-07-26	CUST1116	PROD15	STORE7	3	147.01	0.19	COD	357.2343
SALE6	2025-07-12	CUST1016	PROD56	STORE4	2	785.31	0.15	PayPal	1335.027
SALE7	2023-11-18	CUST1152	PROD70	STORE8	4	342.39	0.23	PayPal	1054.5612
SALE8	2025-05-04	CUST1187	PROD99	STORE5		448.25	0.07	PayPal	1250.6175
SALE9	2025-07-31	CUST1464	PROD33	STORE0	2	873.17	0.23	PayPal	1344.6818
SALE10	2024-02-13	CUST1065	PROD24	STORE14	1	278.63	0.07	Credit Card	259.1259
SALE11	2025-05-24	CUST1320	PROD39	STORE1	4	286.92	0.17	Credit Card	952.5744
SALE12	2024-08-24	CUST1237	PROD88	STORE4	4	644.58	0.01	Credit Card	2552.5368
SALE13	2025-08-06	CUST1443	PROD34	STORE16	2	316.9	0.3	PayPal	443.66
SALE14	2025-06-21	CUST1348	PROD24	STORE15	4	403.47	0.29	PayPal	1145.8548
SALE15	2024-12-10	CUST1435	PROD36	STORE9		871.34	0.04	COD	2509.4592
SALE16	2025-09-13	CUST1276	PROD65	STORE19	2	451.98	0.25	Credit Card	677.97
SALE17	2024-12-11	CUST1079	PROD27	STORE4	2	502.69	0.23	Credit Card	774.1426
SALE18	2024-08-13	CUST1094	PROD39	STORE4	2	559.23	0.25	COD	838.845

2.Data cleaning: find/remove duplicates, handle missing/inconsistent data, correct formatting errors, and validate data

Customer_table

For customer_ID - Replacing the C- with CUST to make it consistent

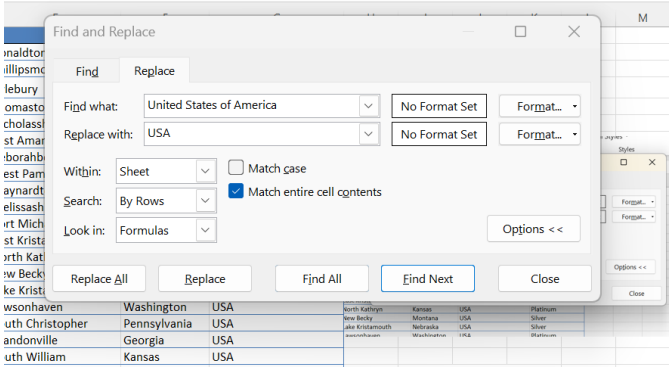


Name - PROPER(TRIM(CLEAN(B2)))

City – PROPER(TRIM(CLEAN(E2)))

State – PROPER(TRIM(CLEAN(F2)))

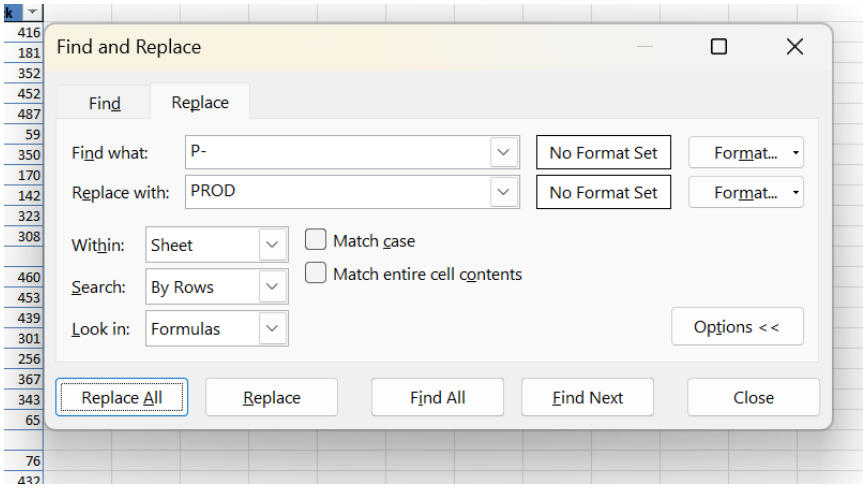
Country – Replacing United States of America to USA



Product_table

For Product Id

Replacing P- with PROD



Cost – IF(ISBLANK([@Cost]),AVERAGE($F$2:$F$101),[@Cost])

Stock – IF(ISBLANK(G2),AVERAGE($G$2:$G$101),G2)

Store_table

There is no inconsistency in text , changed the data type of the columns

Sales_table

Quantity – IF(ISBLANK([@Quantity]),AVERAGE($F$2:$F$2001),[@Quantity])

Unit_Price – IF(ISBLANK([@[Unit_Price]]),AVERAGE($G$2:$G$2001),[@[Unit_Price]])

Discount – IF(ISBLANK([@Discount]),AVERAGE($H$2:$H$2001),[@Discount])

Total_amount – [@[Unit_Price]]*[@Quantity]-[@Discount]

B2 | 07-07-2024

Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
CUST1409	PROD97	STORE3	1	218.04	0.09	Credit Card	217.95
CUST1467	PROD74	STORE17	3	104.47	0.27	PayPal	313.14
CUST1056	PROD49	STORE8	4	914.05	0.25	Credit Card	3655.95
CUST1346	PROD76	STORE8	1	921.51	0.16	COD	921.35
CUST1493	PROD59	STORE5	3	745.30	0.18	Credit Card	2235.72
CUST1116	PROD15	STORE7	3	147.01	0.19	COD	440.84
CUST1016	PROD56	STORE4	2	785.31	0.15	PayPal	1570.47
CUST1152	PROD70	STORE8	4	342.39	0.23	PayPal	1369.33
CUST1187	PROD99	STORE5	2	448.25	0.07	PayPal	1111.68
CUST1464	PROD33	STORE0	2	873.17	0.23	PayPal	1746.11
CUST1065	PROD24	STORE14	1	278.63	0.07	Credit Card	278.56
CUST1320	PROD39	STORE1	4	286.92	0.17	Credit Card	1147.51
CUST1237	PROD88	STORE4	4	644.58	0.01	Credit Card	2578.31
CUST1443	PROD34	STORE16	2	316.90	0.30	PayPal	633.50
CUST1348	PROD24	STORE15	4	403.47	0.29	PayPal	1613.59
CUST1435	PROD36	STORE9	2	871.34	0.04	COD	2161.06
CUST1276	PROD65	STORE19	2	451.98	0.25	Credit Card	903.71
CUST1079	PROD27	STORE4	2	502.69	0.23	Credit Card	1005.15
CUST1094	PROD39	STORE4	2	559.23	0.25	COD	1118.21
CUST1239	PROD60	STORE13	4	513.98	0.04	COD	2055.90
CUST1144	PROD72	STORE17	3	986.04	0.27	COD	2957.85
CUST1396	PROD43	STORE18	1	589.56	0.22	Credit Card	589.34

Quantity	IF(ISBLANK([@Quantity]),AVERAGE($F$2:$F$2001),[@Quantity])
Unit_Price	IF(ISBLANK([@[Unit_Price]]),AVERAGE($G$2:$G$2001),[@[Unit_Price]])
Discount	IF(ISBLANK([@Discount]),AVERAGE($H$2:$H$2001),[@Discount])
Total_amount	[@[Unit_Price]]*[@Quantity]-[@Discount]

merged table | Customer_Dim | Product_Dim | Store_Dim | Sales_Fact

G18

STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
STORE1	Martinez, Adkins and Johnson	North	Danielmouth	Flagship
STORE2	Black-Horn	South	North Nancyview	Outlet
STORE3	Johnson-Wright	North	Hoodberg	Outlet
STORE4	Alexander Group	South	Port Sharon	Online
STORE5	Gray Ltd	North	New Michaelchester	Outlet
STORE6	Benson-Elliott	East	Port Josephberg	Online
STORE7	Martinez-Mullins	South	Williamsfurt	Online
STORE8	Bean, Jones and Park	West	Thompsonborough	Online
STORE9	Moore, Padilla and Butler	South	East Johnstad	Flagship
STORE10	Hill, Wagner and Ballard	East	East Davidville	Online
STORE11	Aguirre-Keller	South	Lake Michael	Online
STORE12	Thomas, Bridges and Harris	North	Sabrinatown	Online
STORE13	Gardner, Humphrey and Contreras	East	Johnsonville	Flagship
STORE14	Mcdonald Inc	South	Christopherland	Outlet
STORE15	Shelton PLC	North	East Jenniferstad	Flagship
STORE16	Frederick-Wright	West	New Jimmymouth	Flagship
STORE17	Holloway Ltd	North	Bensonhaven	Flagship
STORE18	Guerrero-Schmitt	East	East Stevechester	Flagship
STORE19	Mann-Austin	East	Vasquezborough	Online

merged table | Customer_Dim | Product_Dim | Store_Dim | Sales_Fact

Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc	254.04	352
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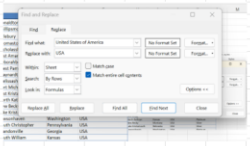
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261.00	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.80	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.40	273
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowman	48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	254.04	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51	273
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodriguez	484.62	432
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos	254.04	360
PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340
PROD25	Women's Scarf Set	Clothing	Accessories	Brown and Sons	174.02	268
PROD26	Wireless Bluetooth Speaker	Electronics	Audio	Reed-Schmidt	322.41	213
PROD27	Decorative Floor Vase	Home	Decor	Khan, Jones and Paul	254.04	378
PROD28	Sports Stopwatch Timer	Sports	Accessories	Burns-Hansen	38.76	137
PROD29	Fabric Sofa Throw Blanket	Home	Decor	Hoffman Group	218.48	276

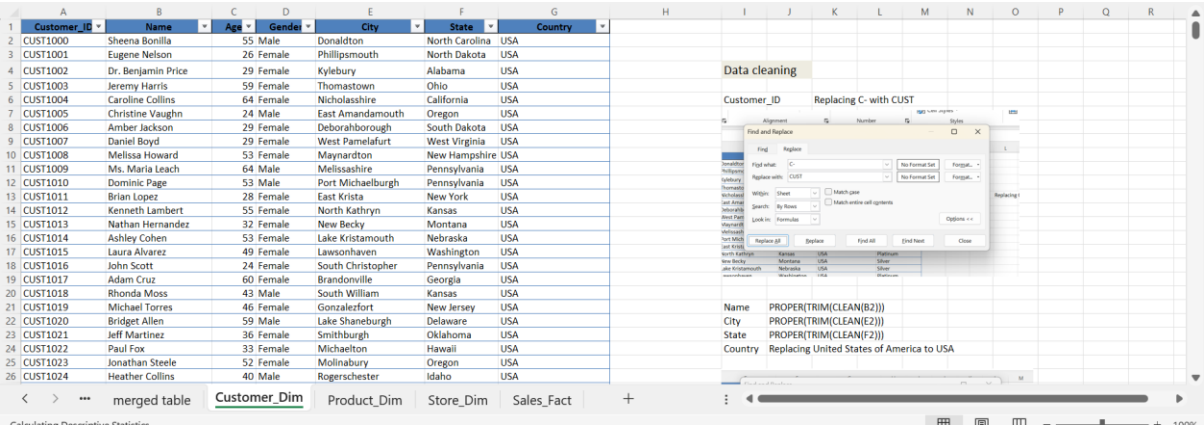
Product_ID	Replacing P- with PROD



Cost	IF(ISBLANK([@Cost]),AVERAGE($F$2:$F$101),[@Cost])
Stock	IF(ISBLANK(G2),AVERAGE($G$2:$G$101),G2)

Sum if the category is electronics	6647.66	SUMIF(Product_table[Category],"Electronics",Product_table[Cost])

merged table | Customer_Dim | Product_Dim | Store_Dim | Sales_Fact



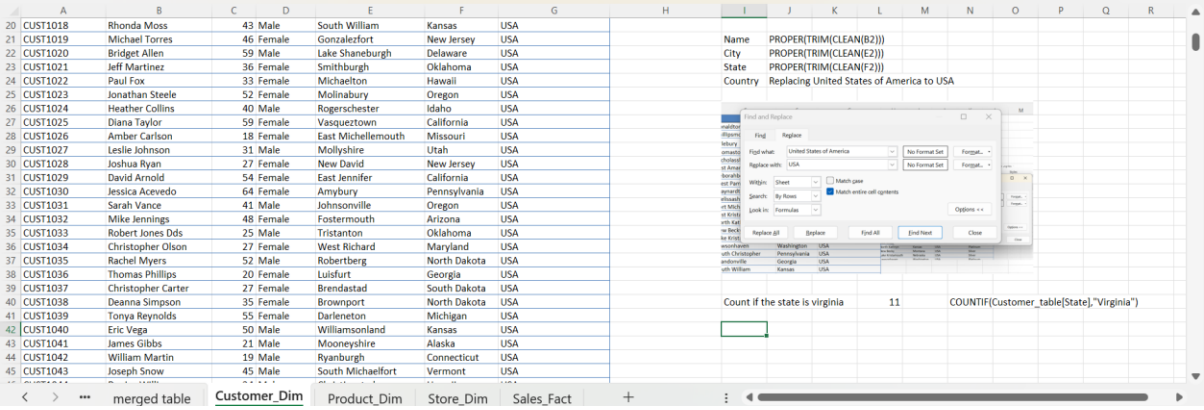
Customer_ID	Name	Age	Gender	City	State	Country
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA

Data cleaning

Customer_ID Replacing C- with CUST

Name PROPER(TRIM(CLEAN(B2)))
City PROPER(TRIM(CLEAN(E2)))
State PROPER(TRIM(CLEAN(F2)))
Country Replacing United States of America to USA

merged table | Customer_Dim | Product_Dim | Store_Dim | Sales_Fact



CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA
CUST1028	Joshua Ryan	27	Female	New David	New Jersey	USA
CUST1029	David Arnold	54	Female	East Jennifer	California	USA
CUST1030	Jessica Acevedo	64	Female	Amybury	Pennsylvania	USA
CUST1031	Sarah Vance	41	Male	Johnsonville	Oregon	USA
CUST1032	Mike Jennings	48	Female	Fostermouth	Arizona	USA
CUST1033	Robert Jones Dds	25	Male	Tristanton	Oklahoma	USA
CUST1034	Christopher Olson	27	Female	West Richard	Maryland	USA
CUST1035	Rachel Myers	52	Male	Robertberg	North Dakota	USA
CUST1036	Thomas Phillips	20	Female	Luisfurt	Georgia	USA
CUST1037	Christopher Carter	27	Female	Brendastad	South Dakota	USA
CUST1038	Deanna Simpson	35	Female	Brownport	North Dakota	USA
CUST1039	Tonya Reynolds	55	Female	Darleneton	Michigan	USA
CUST1040	Eric Vega	50	Male	Williamsonland	Kansas	USA
CUST1041	James Gibbs	21	Male	Mooneyshire	Alaska	USA
CUST1042	William Martin	19	Male	Ryanburgh	Connecticut	USA
CUST1043	Joseph Snow	45	Male	South Michaelfort	Vermont	USA

Name PROPER(TRIM(CLEAN(B2)))
City PROPER(TRIM(CLEAN(E2)))
State PROPER(TRIM(CLEAN(F2)))
Country Replacing United States of America to USA

Count if the state is virginia 11 COUNTIF(Customer_table[State],"Virginia")

merged table | Customer_Dim | Product_Dim | Store_Dim | Sales_Fact

4.Merging all the sheets

Customer name
XLOOKUP([@[Customer_ID]],Customer_table[Customer_ID],Customer_table[Name],FALSE)

Age

XLOOKUP([@[Customer_ID]],Customer_table[Customer_ID],Customer_table[Age],FALSE)

Gender

XLOOKUP([@[Customer_ID]],Customer_table[Customer_ID],Customer_table[Gender],FALSE)

City

XLOOKUP([@[Customer_ID]],Customer_table[Customer_ID],Customer_table[City],FALSE)

State

XLOOKUP([@[Customer_ID]],Customer_table[Customer_ID],Customer_table[State],FALSE)

Country

XLOOKUP([@[Customer_ID]],Customer_table[Customer_ID],Customer_table[Country],FALSE)

Product name

XLOOKUP([@[Product_ID]],Product_table[Product_ID],Product_table[Product_Name],FALSE)

Category

XLOOKUP([@[Product_ID]],Product_table[Product_ID],Product_table[Category],FALSE)

Sub category

XLOOKUP([@[Product_ID]],Product_table[Product_ID],Product_table[Sub_Category],FALSE)

Brand

=XLOOKUP([@[Product_ID]],Product_table[Product_ID],Product_table[Brand],FALSE)

Cost

=XLOOKUP([@[Product_ID]],Product_table[Product_ID],Product_table[Cost],FALSE)

Stock

=XLOOKUP([@[Product_ID]],Product_table[Product_ID],Product_table[Stock],FALSE)

Store name

=XLOOKUP([@[Store_ID]],Store_table[Store_ID],Store_table[Store_Name],FALSE)

Store region

=XLOOKUP([@[Store_ID]],Store_table[Store_ID],Store_table[Region],FALSE)

Store city

=XLOOKUP([@[Store_ID]],Store_table[Store_ID],Store_table[City],FALSE)

Store type

=XLOOKUP([@[Store_ID]],Store_table[Store_ID],Store_table[Store_Type],FALSE)

Z2 =XLOOKUP([@[Store_ID]],Store_table[Store_ID],Store_table[Store_Type],FALSE)

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Customer Name	Age	Gend	
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	218.04	0.09	Credit Card	217.95	Sarah Smith	55	Male	Kathl
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	104.47	0.27	PayPal	313.14	Michael Foley	63	Female	Onei
SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	914.05	0.25	Credit Card	3655.95	Christopher Sutton	49	Female	Debc
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	921.51	0.16	COD	921.35	John Jordan	58	Female	Port
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	745.30	0.18	Credit Card	2235.72	Amanda Powell	28	Male	Soutl
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	147.01	0.19	COD	440.84	Anthony Phillips	18	Female	Nortl
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	785.31	0.15	PayPal	1570.47	John Scott	24	Female	Soutl
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	342.39	0.23	PayPal	1369.33	Patrick Roberts	41	Female	Jarec
SALE8	04-05-2025	CUST1187	PROD99	STORE5	2	448.25	0.07	PayPal	1111.68	Dr. Suzanne Bullock	60	Female	Lake
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	873.17	0.23	PayPal	1746.11	David Pierce	45	Female	Gray
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	278.63	0.07	Credit Card	278.56	Michael Thomas	21	Female	Woo
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	286.92	0.17	Credit Card	1147.51	Jessica Gonzalez	34	Female	Soutl
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	644.58	0.01	Credit Card	2578.31	Wesley Short	41	Female	Susa
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	316.90	0.30	PayPal	633.50	Adam Castro	51	Male	New
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	403.47	0.29	PayPal	1613.59	Sherry Cabrera	55	Male	Shan
SALE15	10-12-2024	CUST1435	PROD36	STORE9	2	871.34	0.04	COD	2161.06	Lindsay Beasley	37	Female	Wesl
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	451.98	0.25	Credit Card	903.71	Arthur Warren	38	Male	Ericsl
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	502.69	0.23	Credit Card	1005.15	Molly Gardner	22	Female	Nicol
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	559.23	0.25	COD	1118.21	David Patterson	43	Female	New
SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	513.98	0.04	COD	2055.90	Maria Hill	58	Female	Zam
SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	986.04	0.27	COD	2957.85	Justin Cross	62	Female	Nortl

merged table | Customer_Dim | Product_Dim | Store_Dim | Sales_Fact | Customer

Ready Accessibility: Investigate 89%

Z2 =XLOOKUP([@[Store_ID]],Store_table[Store_ID],Store_table[Store_Type],FALSE)

Customer city	Customer state	Count	Product name	Product category	Sub category	Brand	cost	Stoc	Store name
Kathleenport	Wyoming	USA	Cricket Bat	Sports	Team Sports	Khan LLC	245.98	471	Johnson-Wright
Oneillfurt	Vermont	USA	Bluetooth Car Audio System	Electronics	Audio	Bruce-Bowen	403.71	159	Holloway Ltd
Deborahberg	Pennsylvania	USA	Men's Roll-On Perfume	Beauty	Fragrance	Sullivan and Sons	43.29	267	Bean, Jones and Park
Port Crystalberg	Tennessee	USA	Wireless Mouse	Electronics	Computer	Larson Ltd	392.89	320	Bean, Jones and Park
South William	Alabama	USA	Men's Hair Styling Gel	Beauty	Haircare	Cline-Mccarthy	17.31	399	Gray Ltd
North Davidport	Rhode Island	USA	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301	Martinez-Mullins
South Christopher	Pennsylvania	USA	Premium Blanket Throw	Home	Decor	Hernandez, Kennedy and Porter	461.47	190	Alexander Group
Jaredville	Missouri	USA	Kitchen Storage Rack	Home	Storage	Duran Group	138.05	89	Bean, Jones and Park
Lake Blaketon	Alabama	USA	Stainless Steel Cookware Set	Home	Kitchen	Campbell Ltd	201.41	146	Gray Ltd
Grayland	Washington	USA	Modern Wall Clock	Home	Decor	Weiss-Watson	179.77	305	Rivera-Warner
Woodmouth	Nevada	USA	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340	Mcdonald Inc
South Darrenland	Iowa	USA	Portable Power Bank	Electronics	Mobile	Cummings and Sons	326.06	301	Martinez, Adkins and Johnson
Susanstad	Montana	USA	Sports Wrist Sweatbands	Sports	Accessories	Olsen-Daniels	131.24	232	Alexander Group
New Donnabury	Texas	USA	Gaming Mechanical Keyboard	Electronics	Computer	Lucero LLC	419.28	86	Frederick-Wright
Shannonview	Pennsylvania	USA	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340	Shelton PLC
Wesleyhaven	Idaho	USA	Women's Summer Dress	Clothing	Tops	Smith-Mills	223.43	154	Moore, Padilla and Butler
Ericshire	Nevada	USA	Adjustable Dumbbells 20kg	Sports	Fitness	Montgomery, Benton and Miller	240.63	423	Mann-Austin
Nicoleborough	Arkansas	USA	Decorative Floor Vase	Home	Decor	Khan, Jones and Paul	254.04	378	Alexander Group
New Amyport	Nevada	USA	Portable Power Bank	Electronics	Mobile	Cummings and Sons	326.06	301	Alexander Group
Zamoraborough	Nevada	USA	Luxury Perfume Spray	Beauty	Fragrance	Gibson, Green and Lopez	482.75	444	Gardner, Humphrey and Contrera
North Anthonychester	Nevada	USA	Professional Jump Rope	Sports	Fitness	Davis PLC	477.61	300	Holloway Ltd

merged table | Customer_Dim | Product_Dim | Store_Dim | Sales_Fact | Customer

Z2 =XLOOKUP([@[Store_ID]],Store_table[Store_ID],Store_table[Store_Type],FALSE)

Brand	cost	Stoc	Store name	Store region	Store city	store_type
Khan LLC	245.98	471	Johnson-Wright	North	Hoodberg	Outlet
Bruce-Bowen	403.71	159	Holloway Ltd	North	Bensonhaven	Flagship
Sullivan and Sons	43.29	267	Bean, Jones and Park	West	Thompsonborough	Online
Larson Ltd	392.89	320	Bean, Jones and Park	West	Thompsonborough	Online
Cline-Mccarthy	17.31	399	Gray Ltd	North	New Michaelchester	Outlet
Thomas LLC	465.34	301	Martinez-Mullins	South	Williamsfurt	Online
Hernandez, Kennedy and Porter	461.47	190	Alexander Group	South	Port Sharon	Online
Duran Group	138.05	89	Bean, Jones and Park	West	Thompsonborough	Online
Campbell Ltd	201.41	146	Gray Ltd	North	New Michaelchester	Outlet
Weiss-Watson	179.77	305	Rivera-Warner	East	Washingtonfurt	Flagship
Cook Ltd	166.06	340	Mcdonald Inc	South	Christopherland	Outlet
Cummings and Sons	326.06	301	Martinez, Adkins and Johnson	North	Danielmouth	Flagship
Olsen-Daniels	131.24	232	Alexander Group	South	Port Sharon	Online
Lucero LLC	419.28	86	Frederick-Wright	West	New Jimmymouth	Flagship
Cook Ltd	166.06	340	Shelton PLC	North	East Jenniferstad	Flagship
Smith-Mills	223.43	154	Moore, Padilla and Butler	South	East Johnstad	Flagship
Montgomery, Benton and Miller	240.63	423	Mann-Austin	East	Vasquezborough	Online
Khan, Jones and Paul	254.04	378	Alexander Group	South	Port Sharon	Online
Cummings and Sons	326.06	301	Alexander Group	South	Port Sharon	Online
Gibson, Green and Lopez	482.75	444	Gardner, Humphrey and Contreras	East	Johnsonville	Flagship
Davis PLC	477.61	300	Holloway Ltd	North	Bensonhaven	Flagship

merged table | Customer_Dim | Product_Dim | Store_Dim | Sales_Fact | Customer

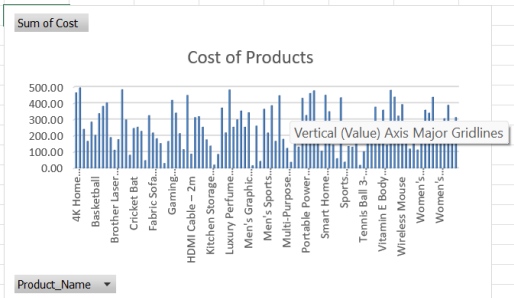
5.Pivot tables created based on questions

Product name	Sum of Cost
4K Home Projector	465.34
Adjustable Dumbbell Set	495.72
Adjustable Dumbbells 20kg	240.63
Adjustable Jump Rope	167.12
Badminton Racket Set	285.54
Basketball	204.39
Beard Grooming Kit	337.42
Bedroom Table Lamp	382.22
Bluetooth Car Audio System	403.71
Boxing Hand Wraps	190.29
Brother Laser Printer	112.42
Car Dashboard Phone Holder	177.98
Carbon Fiber Tennis Racket	484.62
Ceramic Kitchen Bowl Set	298.86
Ceramic Plant Pot	81.75
Cricket Bat	245.98
Decorative Floor Vase	254.04
Decorative Lamp Shade	228.53
Decorative Wall Mirror	48.01
Essential Oil Diffuser	325.26
Fabric Sofa Throw Blanket	218.48

Minimum cost	17.31
Maximum cost	495.72
Product with minimum cost	Men's Hair Styling Gel
Product with maximum cost	Adjustable Dumbbell Set

XLOOKUP(G4,B4:B103,A4:A103,FALSE)

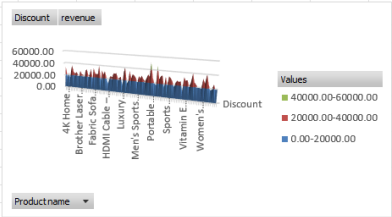
XLOOKUP(G5,B4:B103,A4:A103,FALSE)



Product vs discount | product vs cost | merged table | Customer_Dim | Product_Dir

Product	Discount	revenue
4K Home Projector	4.25	33326.81
Adjustable Dumbbell Set	2.48	15735.20
Adjustable Dumbbells 20kg	2.28	22755.49
Adjustable Jump Rope	1.65	15878.65
Badminton Racket Set	4.08	27320.02
Basketball	3.40	36054.24
Beard Grooming Kit	2.81	25212.02
Bedroom Table Lamp	2.83	22208.58
Bluetooth Car Audio System	3.77	25743.67
Boxing Hand Wraps	2.92	22231.57
Brother Laser Printer	2.21	19088.52
Car Dashboard Phone Holder	3.87	28027.71
Carbon Fiber Tennis Racket	1.83	22732.58
Ceramic Kitchen Bowl Set	3.42	21977.30
Ceramic Plant Pot	2.10	18839.73
Cricket Bat	3.06	29227.85
Decorative Floor Vase	3.99	31591.73
Decorative Lamp Shade	2.83	19702.70
Decorative Wall Mirror	1.95	24232.47
Essential Oil Diffuser	3.25	32394.13
Fabric Sofa Throw Blanket	2.83	15773.14
Face Foundation Kit	4.80	34687.34
Fitness Resistance Bands	1.93	26072.42
Foam Roller	2.79	25234.74
Foldable Camping Chair	2.95	25646.20
Gaming Mechanical Keyboard	3.95	39584.37

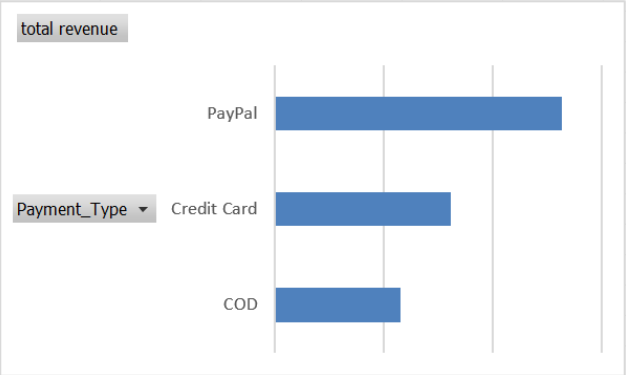
Descriptive analytics	
Mean	5.864799
Standard Error	2.904038
Median	2.86
Mode	2.85
Standard Deviation	29.18522
Sample Variance	851.7771
Kurtosis	100.8621
Skewness	10.03969
Range	294.5623
Minimum	1.61
Maximum	296.1723
Sum	592.3447
Count	101
Largest(1)	296.1723
Smallest(1)	1.61
Confidence Level(95.0%)	5.761529



Product vs discount | product vs cost | merged table | Customer Dim | Product Dir

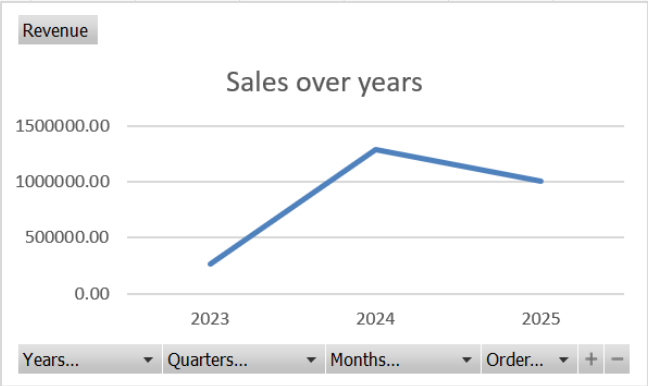
Payment type	total revenue
COD	826286.17
Credit Card	844455.89
PayPal	885516.27
Grand Total	**2556258.32**

Most revenue is received through PayPal



payment type vs sales | Product vs discount | product vs cost | merged table | Cust

Order year	Revenue
2023	264985.50
2024	1286258.95
2025	1005013.87
Grand Total	**2556258.32**



year vs revenue | payment type vs sales | Product vs discount | product vs cost | m

store name	sales in each store	Quantity sold
Aguirre-Keller	109128.64	206
Alexander Group	144009.60	255
Bean, Jones and Park	185351.24	344
Benson-Elliott	134996.75	258
Black-Horn	116155.25	220
Frederick-Wright	131216.55	249
Gardner, Humphrey and Contreras	137594.27	243
Gray Ltd	96029.86	201
Guerrero-Schmitt	123258.38	247
Hill, Wagner and Ballard	129840.12	255
Holloway Ltd	145111.41	276
Johnson-Wright	148179.71	293
Mann-Austin	109963.71	240
Martinez, Adkins and Johnson	115885.44	231
Martinez-Mullins	125499.18	273
Mcdonald Inc	122885.99	224
Moore, Padilla and Butler	127839.74	237

store vs sales | year vs revenue | payment type vs sales | Product vs discount | proc

Customer names	sales amount
Aaron Luna	4582.52
Adam Castro	8182.67
Adam Cruz	4990.31
Adam Rowland	4966.34
Alejandro Doyle	5781.18
Alejandro Evans	2782.16
Alex Mann Ii	3901.58
Alexander Byrd	5728.75
Alexis Castaneda	5358.93
Alyssa Vargas	7431.59
Amanda Powell	5643.86
Amber Carlson	3638.86
Amber Jackson	7846.33
Amber Mcclain	15913.16
Amy Fowler	5425.23
Amy Walker	8575.03
Amy Wilson	11228.35
Andrea Harris	10162.71
Andrea Jacobs	3902.95
Andrea Perez	2159.82
Andrew Mason	7608.50
Andrew Smith	10527.74
Angela Robinson	3408.61

Customer name who generates highest revenue: Jordan Wells

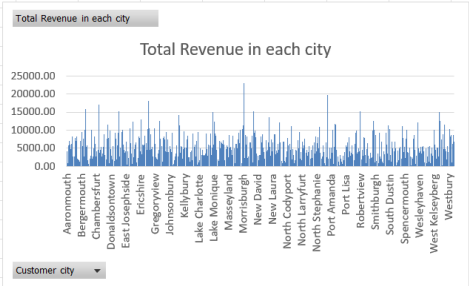
XLOOKUP(MAX(B4:B493),B4:B493,A4:A493,FALSE)

Customer vs gender | city vs sales | Customer vs sales

Cities	Total Revenue in each city
Aaronmouth	4195.38
Aguilarland	992.24
Aliciahaven	5838.46
Amandaborough	6962.64
Amandaside	6085.29
Amberburgh	4769.84
Amybury	6225.31
Anabury	8252.65
Andersonside	2713.75
Andrewsfurt	2771.22
Andrewsland	7791.56
Anitaland	7046.99
Anthonymouth	8255.61
Ashleymouth	2088.98
Ashleystad	1954.04
Austinfurt	1678.40
Avilahaven	1222.28
Benjaminton	7029.39
Bergermouth	9499.49

Descriptive Analysis using tool pak

Mean	5370.290592
Standard E	155.6557249
Median	4860.33
Mode	#N/A
Standard D	3396.006983
Sample Va	11532863.43
Kurtosis	2.33384478
Skewness	1.17099025
Range	22946.68
Minimum	40.82
Maximum	22987.5
Sum	2556258.322
Count	476
Largest(1)	22987.5
Smallest(1	40.82
Confidenc	305.8589501

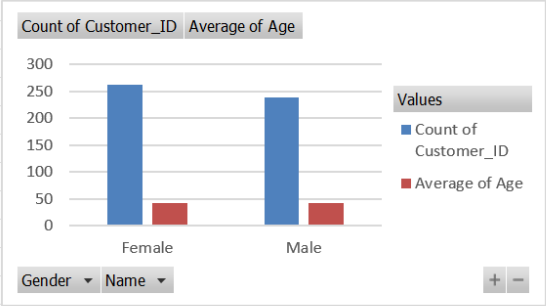


Customer vs gender | city vs sales | Customer vs sales | region vs store | store vs sa

Gender	Count of Customer_ID	Average of Age
Female	262	42
Male	238	41
Grand Total	500	42

Insights

Female customers are more than male



Customer vs gender | city vs sales | Customer vs sales | region vs store | store vs sa

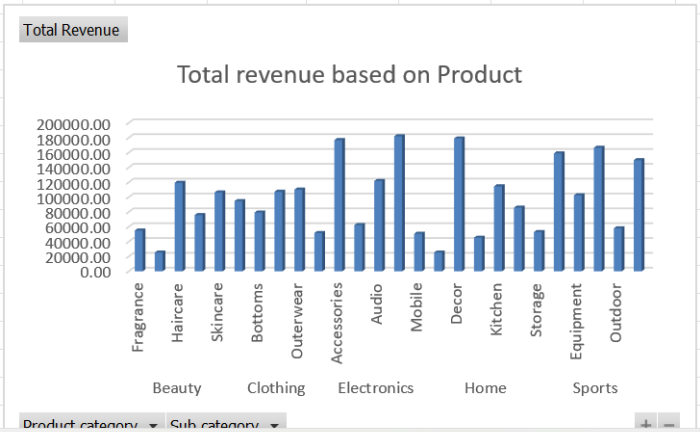
Product ID	Number of sales	Total Quantity got sold	Total Revenue for each product
PROD0	16	39	19088.52
PROD1	15	41	19261.56
PROD10	21	51	21977.30
PROD11	19	53	31275.61
PROD12	19	41	25788.17
PROD13	22	55	25164.41
PROD14	18	45	24232.47
PROD15	29	72	33326.81
PROD16	22	62	25212.02
PROD17	29	70	32666.31
PROD18	16	37	18197.85
PROD19	27	63	28789.80
PROD2	16	41	19953.62
PROD20	20	53	32139.99
PROD21	22	42	23879.80
PROD22	19	43	22732.58
PROD23	15	34	14889.12
PROD24	19	48	25646.20
PROD25	23	58	26804.04
PROD26	20	49	25816.77
PROD27	21	56	31591.73
PROD28	18	43	18211.88

Maximum sales happen in	PROD39	XLOOKUP(MAX(B4:B103),B4:B103,A4:A103,FALSE)
Minimum sales happen in	PROD32	XLOOKUP(MIN(B4:B103),B4:B103,A4:A103,FALSE)
Highest quantity sold in product	PROD37	XLOOKUP(MAX(C4:C103),C4:C103,A4:A103,FALSE)
Minimum quantity sold in product	PROD32	XLOOKUP(Min(C4:C103),C4:C103,A4:A103,FALSE)
		XLOOKUP(MAX(D4:D103),D4:D103,A4:A103,FALSE)

product vs sales | Customer vs gender | city vs sales | Customer vs sales | region vs

Category	Total Revenue
Beauty	**381526.08**
Fragrance	54923.20
Grooming	25212.02
Haircare	119336.10
Makeup	75741.81
Skincare	106312.95
Clothing	**442751.50**
Accessories	94596.46
Bottoms	79182.55
Footwear	107211.91
Outerwear	110137.71
Tops	51622.87
Electronics	**593563.45**
Accessories	176880.50
Appliances	62386.75
Audio	121741.16
Computer	181939.69
Mobile	50615.34
Home	**502977.27**

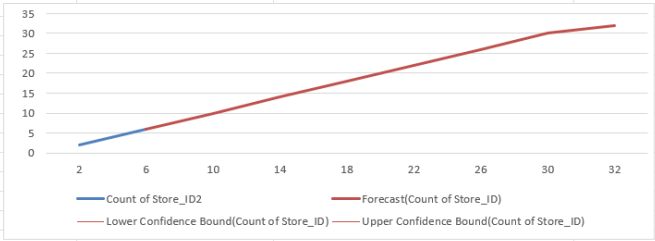
Maximum revenue	381526.08	Sports
Mininum revenue	25212.02	Beauty

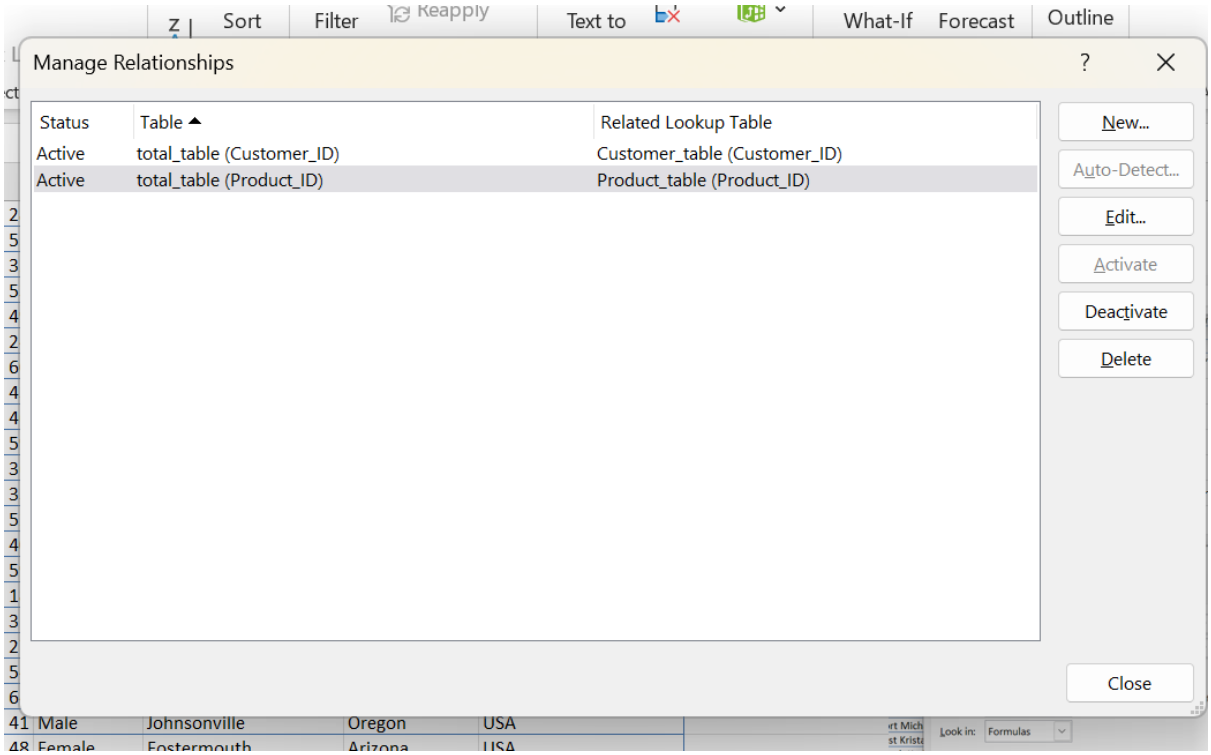


Category vs sales | product vs sales | Customer vs gender | city vs sales | Customer

Sample for forecast

Count of Store_ID	Count of Store_ID2	Forecast(Count of Store_ID)	Lower Confidence Bound(Count of Store_ID)	Upper Confidence Bound(Count of Store_ID)
2	2			
6	6	6	6.00	6.00
10		10	10.00	10.00
14		14	14.00	14.00
18		18	18.00	18.00
22		22	22.00	22.00
26		26	26.00	26.00
30		30	30.00	30.00
32		32	32.00	32.00



Forecast for region vs store | Category vs sales | product vs sales | Customer vs gend

Sample relationship